

Ashwary Banik Dolan

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EDUCATION

Masters of Science in Hydrogeology and Water Resources Management <i>University of Dhaka(DU)</i>	Jan 2024 – Jun 2026 <i>Dhaka, Bangladesh</i>
Bachelor of Science in Geology <i>University of Dhaka(DU)</i> CGPA – 3.33/4.00	Jan 2019- Nov 2023 <i>Dhaka, Bangladesh</i>

RESEARCH EXPERIENCE

Research Assistant <i>Groundwater Modeling Lab, Department of Geology, DU</i>	Jun 2025 – Present
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Title: Groundwater Monitoring in Camps and Surrounding Host Communities, Ukhiya.

- Goals and Vision:** Investigate the feasibility of deeper groundwater reserves or surface water sources to diversify and secure long-term water supply for the camps.

Remote Research Collaborator	Aug 2025 – Present
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Title: Post wildfire ground movement characterization and mapping

Supervisor: *Kshama Roy, Assistant Professor, Department of Civil Engineering, Memorial University*

- Goals and Vision:** Characterize Post Wildfire Landslide Mechanisms. Utilize remote sensing and geospatial analysis to establish baseline landslide susceptibility models in burned areas, validated through historical data.

Postgraduate Thesis	Jan 2024 – Dec 2024
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Title: Evaluation of Groundwater Sustainability in watershed of Dhaka by Constructing GIS Based Groundwater Budget

Supervisor: *Dr. Md. Mahfuzur R Khan, Associate Professor, Department of Geology, DU*

- Climate & hydrologic data pre-processing in ArcGIS Pro and Python.
- Hyperparameter-tuned Random Forest for detailed spatiotemporal LULC change analysis.
- WetSpss based water budget recharge estimation and quantify LULC-driven temporal shifts in net recharge.

FIELD RESEARCH

Integrated multidisciplinary field work on Coastal Geomorphology (Cox's Bazar, Bangladesh)	Jan 2023
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(Key Skills: Spatial Data Analysis, Water Quality Testing, Hydrogeochemical Analysis, Geophysical Survey)

- Performed VES and ERT geophysical surveys to delineate and track saline freshwater interfaces.
- Utilized major-ion chemistry, statistical analysis and hydrochemical diagrams to identify dominant geochemical processes such as weathering and cation exchange, trace salinity sources.

Geological field work on stream and road-cut sections (Jaflong, Sylhet, Bangladesh)	Jan 2022
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(Key Skills: Sedimentary sequences, Sediment Core Collection, Seismic Interpretation, Geochemical Analysis)

- Mapped Eocene–Miocene stratigraphy and refined the Sylhet Trough basin framework.
- Identified economic units (limestone) and screened hydrocarbon potential.
- Analyzed Dauki Fault Zone controls on petroleum system.

Geological field work on Lichubagan-Kaptai road-cut sections (Rangamati, Bangladesh)	Nov 2021
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(Key Skills: Stratigraphic Logging, Petrographic Analysis, Erosion and Deposition Studies, Topographic Surveying)

- Characterized an asymmetric fold (steeper west flank) and mapped associated thrust faults.
- Collected and Logged stratigraphy from Bhuban to Dupi Tila and documented sedimentary structures

Geological field mapping on Sitakund Anticline (Chittagong, Bangladesh) Jan 2020
(Key Skills: Structural Geology Analysis, Mineral Identification, Geological Mapping (GIS, GPS), Rock Sampling)

- Mapped sedimentary lithology and key structures folds, faults, unconformities.
- Documented gas seepages, potential hydrocarbon presence.
- Analyzed the overturned fold at Barabkund, strengthening the structural model and interpretation.

PROFESSIONAL EXPERIENCE

Research Consultant Apr 2025 – June 2025
Center for Environmental and Geographic Information Services (CEGIS)

- Interpreted high-resolution multispectral satellite imagery to extract detailed, plot-level Land Use Land Cover classifications comprising 103 categories.
- Analyzed time series data from Sentinel-2 imagery to identify crop patterns and assess crop intensity.

GIS Analyst Jan 2025 – Apr 2025
GMGI Solutions Limited

- Supported the Cox's Bazar District Master Plan by assembling production-grade GIS datasets and producing basemaps, infrastructure inventories, LULC change layers, and multi-hazard maps (flood, erosion, landslide).
- Delivered stakeholder-ready dashboards and web maps for review, scenario exploration, and decision support.

TECHNICAL SKILLS

Hydrologic Modeling Packages: HEC-HMS, SWAT, MODFLOW, WetSpa
GIS Software: ArcGIS Pro, ArcGIS online, Google Earth Engine
Remote Sensing Software: ENVI, ERDAS IMAGINE, SNAP
Geophysical Techniques: VSE, ERT
Visualization Software: Power BI, Mapbox
Programming Languages: Python, R, JavaScript
Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, GeoPandas,
Adobe Applications: Illustrator, Photoshop, Lightroom
Languages: Fluent in English and Bengali

CERTIFICATES

Spatial Data Science Oct 2024
Esri
Machine Learning with Python July 2024
DataCamp | Credential ID: #33,724,660
Deep Learning Specialization May 2024
DeepLearning.AI | Coursera

EXTRACURRICULAR EXPERIENCES

Association for Women geoscientist Dhaka University student chapter Nov 2023 – Present
Student Affiliated Member
Geo Photographic Club, University of Dhaka Feb 2021 – May 2023
General Secretary

REFERENCE

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